



ISLAMIC UNIVERSITY OF TECHNOLOGY



Course Outline and Course Plan (MATH 4141)

Academic Session: 2024-2025

Name of the Teacher		Dr. Muntasir Alam	Position	Assistant Professor (PT)
Department		Computer Science and Engineering	Program	B. Sc. in SWE
Course Code		MATH 4141	Course Title	Geometry and Differential Calculus
Academic Year		2024-2025	Semester	First
Contact Hours		4.0	Credit Hours	4.0
Textbooks and Authors of the Reference books		1. Pre-calculus, 10th Edition; <i>Michael Sullivan</i>. 2. Calculus Early Transcendental, 10th Edition; <i>Howard Anton, Irl Bivens and Stephen Davis</i>. 3. Calculus Early Transcendental, 8th Edition; <i>James Stewart</i>. 4. Calculus with Analytic Geometry, 2nd Edition; <i>George F. Simmons</i>. 5. Calculus with Analytic Geometry; <i>Alternate Edition, Earl W. Swokowski</i>. 6. Pre-calculus with calculus previews; <i>D. G. Zill and J. M. Dewar</i>.		
Prerequisites		N/A		
Curriculum Requirement		Compulsory		
Course Homepage				
Teaching Methods/ Approaches	☒ Lecture	☐ Group discussion	☒ Demonstration	☒ Problem solving
	☐ Project	☐ Others: Presentation by Students		
Teaching aids	☒ Multi-media	☐ OHP	☒ Board and Marker	☐ Others

Course Assessment Method							
Quiz 20% of the Total Marks (4*20=80 Marks)						Mid Semester (40%)	Semester Final (40%)
1 st Quiz	2 nd Quiz	3 rd Quiz	4 th Quiz	Others		Week /Date	Week /Date
Week /Date	Week /Date	Week /Date	Week /Date	Assignment	Homework		
3 rd Week	6 th Week	10 th Week	13 th Week	Will be given accordingly	Will be given accordingly	As per the schedule of IUT	As per the schedule of IUT

Course Contents and Objectives	Course Contents
	2D Coordinate Geometry: Change of axes: Transform of coordinates, Simplification of equations of curves; Pair of straight lines: Homogeneous second-degree equations, Conditions for general second-degree equations to represent a pair of straight lines, Angle between the lines, Pair of straight lines joining the origin to the points of intersection of the curve and a line, Circles and system of circles: Tangent and normal, Radial axis and its properties, Parametric coordinates.
	3D Coordinate Geometry: Rectangular coordinates, Direction cosines and angle between two lines, The plane and the straight lines, The equation of a sphere, The standard forms of equations of the central conicoid, cones and cylinders.
	Differential Calculus: Limits, Continuity and Differentiability; Differentiation of explicit and implicit functions and parametric equations; Significance of derivatives, Differentials; Successive differentiation of various types of functions; Leibnitz's Theorem; Rolle's Theorem; Mean Value Theorem; Taylor's Theorem in finite and infinite forms; Maclaurin's Theorem in finite and infinite forms; Lagrange's form of remainders; Cauchy's form of remainders; Expansion of functions by differentiation and integration; Partial differentiation; Euler's Theorem; Tangent, maximum and minimum values of functions and points of inflection; Applications of Differential Calculus; Evaluation of indeterminate forms by L'Hospitals rule; Curvature, the center

	of curvature and chord of curvature; Evolutes and involutes; Asymptotes, Envelopes, Curve tracing. <u>Objectives</u> ➤ The geometry part of this course includes an in-depth analysis of plane, solid, and coordinate geometry as they relate to both abstract mathematical concepts as well as real-world problems. ➤ From this course, students will deepen their understanding of geometric relationships in a plane in space, the meaning, and nature of proofs, deductive proof in both mathematical and non-mathematical situations, and integrate geometry with arithmetic and vector algebra. ➤ Emphasis will be placed on developing critical thinking skills as they relate to logical reasoning and arguments. ➤ This course aims to provide a firm foundation in the concepts and techniques of differential calculus, including basic functions, graphs, and their properties, curve sketching, limits, continuity, differentiation, relative extrema, and its applications, etc. ➤ Helps students acquire several techniques of differentiation alongside its applications in real life.
Course Outcomes	<u>Course Outcomes (COs)</u> CO1 – Constructing the definition and theories of well-known mathematical concepts along with necessary conditions or properties. CO2 – Applying the fundamental theories and models of coordinate geometry and differential calculus in real-life applications especially encountered in the fields of science and engineering. CO3 – Representing real-life applications employing the fundamental tools of analytic geometry and differential calculus. CO4 – Solving and modifying the real-life oriented applications by means of the well-known techniques of computational mathematics.

Weekly Plan for Covering the Course Contents

Weeks	Topics	Task/Reading
1	Transformation of coordinates.	
2	Simplification of equations of curves.	
3	Homogeneous second-degree equations, Conditions for general second-degree equations to represent a pair of straight lines, and the angle between the lines.	Quiz 1 – CO1, CO2, CO3, CO4.
4	Pair of straight lines joining the origin to the points of intersection of the curve and a line, Circles and system of circles: Tangent and normal, Radial axis and its properties, Parametric coordinates.	
5	Rectangular coordinates, Direction cosines, and angle between two lines.	
6	The plane and the straight lines; The equation of a sphere.	Quiz 2 – CO1, CO2, CO3, CO4.
7	The standard forms of equations of the central conicoid, cones, and cylinders.	
	MID-Semester Examination	MID - CO1, CO2, CO3, CO4.
8	Limits, Continuity, and Differentiability; Differentiation of explicit and implicit functions and parametric equations;	
9	Significance of derivatives; Differentials; Successive differentiation of various types of functions; Leibnitz's Theorem; Rolle's Theorem; Mean Value Theorem.	
10	Applications of Differential Calculus; Evaluation of indeterminate forms by L'Hospitals rule.	Quiz 3 – CO1, CO2, CO3, CO4.

11	Tangent, maximum and minimum values of functions, and points of inflection; Partial differentiation; Euler's Theorem.	
12	Taylor's Theorem in finite and infinite forms; Maclaurin's Theorem in finite and infinite forms; Lagrange's form of remainders; Cauchy's form of remainders.	
13	Expansion of functions by differentiation and integration.	Quiz 4 – CO1, CO2, CO3, CO4.
14	Curvature, the center of curvature and chord of curvature, Evolutes, and involutes, Asymptotes, envelopes, curve tracing.	
15	Review Class	
	Semester Final Examination	Final - CO1, CO2, CO3, CO4.

Mapping of Course Outcomes (COs) and Evaluation Methods

Assessment Method	Marks	CO1	CO2	CO3	CO4	Final
		PO1	PO1/PO2	PO1/PO2	PO1/PO2	
Class test/Quiz I, Quiz II/Quiz III/Quiz IV	20% (80)	7%	6%	5%	2%	Sum will be counted
Attendance (Class Participation)	Eligibility to sit for the final exam of the course requires 85% attendance.					
Midterm Exam.	40% (160)	15%	15%	5%	5%	As obtained
Final Exam.	40% (160)	5%	5%	15%	15%	As obtained
Total	100% (400)					As obtained

Student Learning Time

NO.	Teaching and Learning Activities	Student Learning Time (SLT): Notional Hours
1	Lectures	60 hours
2	Tutorial	4 hours
4	Assignment/Problem solving/Tests	16 hours
5	Personal study	70 hours
6	Cooperative learning	10 hours
TOTAL SLT		160 hours
CREDIT = SLT/40		4

CO-PO Mapping with Bloom's Taxonomy

Mapping with CO, PO and Bloom's Taxonomy			
CO No.	Course Outcomes (CO) Statement	Levels of Bloom's Taxonomy	Matching with Program Outcome (PO)
CO1	Constructing the definition and theories of well-known mathematical concepts along with necessary conditions or properties.	C4	PO1
CO2	Applying the fundamental theories and models of coordinate geometry and differential calculus in real-life applications especially encountered in the fields of science and engineering.	C3	PO2
CO3	Representing real-life applications employing the fundamental tools of analytic geometry and differential calculus.	C3	PO3

CO4	Solving and modifying the real-life oriented applications by means of the well-known techniques of computational mathematics.	C3	PO4
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Program Outcomes (POs: PO1 ~ PO12)

PO No.	Program Outcomes (POs)
	Students graduating from the Bachelor of Science in Computer Science and Engineering (B.Sc. in CSE) program, upon graduation students will have the ability to:
PO1	Engineering knowledge: Apply knowledge of mathematics, natural science, engineering fundamentals, and an engineering specialization as specified in K1 to K4 respectively to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using the first principles of mathematics, natural sciences, and engineering sciences. (K1 to K4)
PO3	Design/development of solutions: Design solutions for complex engineering problems and design systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. (K5)
PO4	Investigation: Conduct investigations of complex problems using research-based knowledge (K8) and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PO5	Modern tool usage: Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an understanding of the limitations. (K6)
PO6	The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities

	relevant to professional engineering practice and solutions to complex engineering problems. (K7)
PO7	Environment and sustainability: Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts. (K7)
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice. (K7)
PO9	Individual work and teamwork: Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Grading Policy

Numeric Grade	Letter Grade	Grade Point
80% and above	A+	4.00
75% to less than 80%	A	3.75
70% to less than 75%	A-	3.50
65% to less than 70%	B+	3.25

60% to less than 65%	B	3.00
55% to less than 60%	B-	2.75
50% to less than 55%	C+	2.50
45% to less than 50%	C	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00
Satisfactory	S	
Unsatisfactory	U	
Incomplete	I	
Withdrawn	W	

Class Schedule

Section	Day	Time	Room Number
SWE Section 1 (SW1)	Tuesday (MUA)	08:00 – 09:40	301 (AB2)
SWE Section 1 (SW1)	Friday (MUA)	10:30 – 12:10	301 (AB2)

Student's Consulting Hours: Thursday: 09:40-10:30 & Friday: 10:00 -10:30

Contact Details:

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